

Self-Driving RC Car

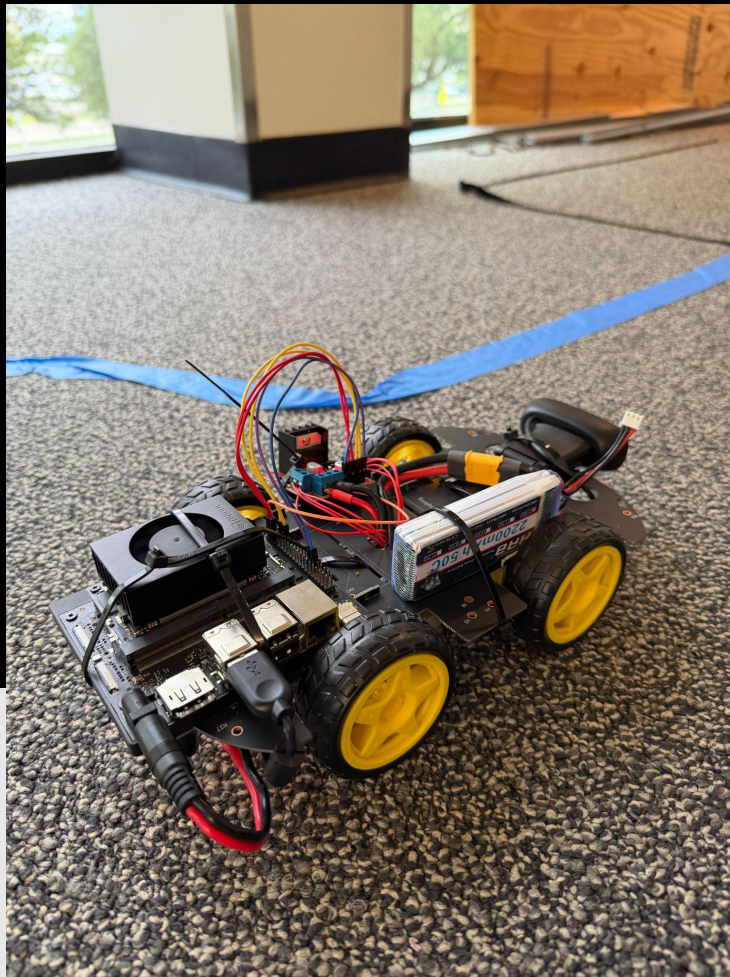
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Computer Vision Project

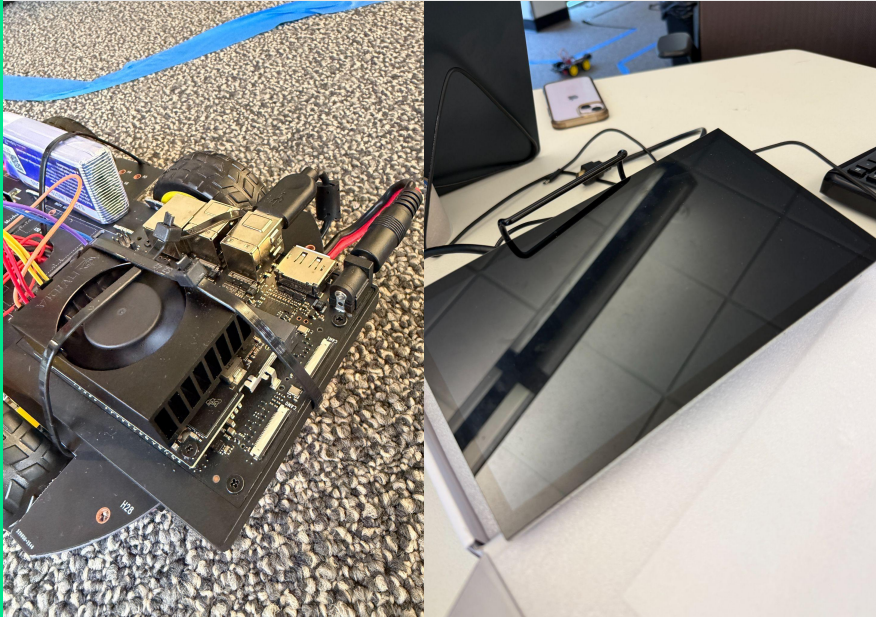


Intro/Goal

- Goal: Construct an RC Car capable of self driving using camera based lane detection and obstacle avoidance
- Computer Vision allows the car to detect and stay within driving lanes as well as avoiding objects
- Mimic already utilized driver assistance technology in newer real world vehicles at a smaller scale



Hardware Setup



- XiaoR 1/10 scale 4WD RC Chassis
- NVIDIA Jetson Orin Nano
- L298N Motor Driver
- Step-Up module
- 2200mAh 7.4V LiPo Battery
- Logitech C270 USB camera
- HamtySan 10.1 Monitor

Software Setup

Main Model

DeepLab V3 ; MobileNetV3
- semantic segmentation

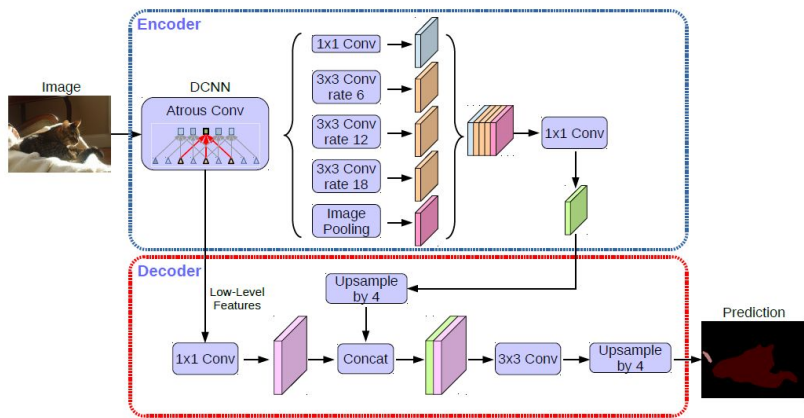
Libraries/Tools

PyTorch / Torchvision - model handling

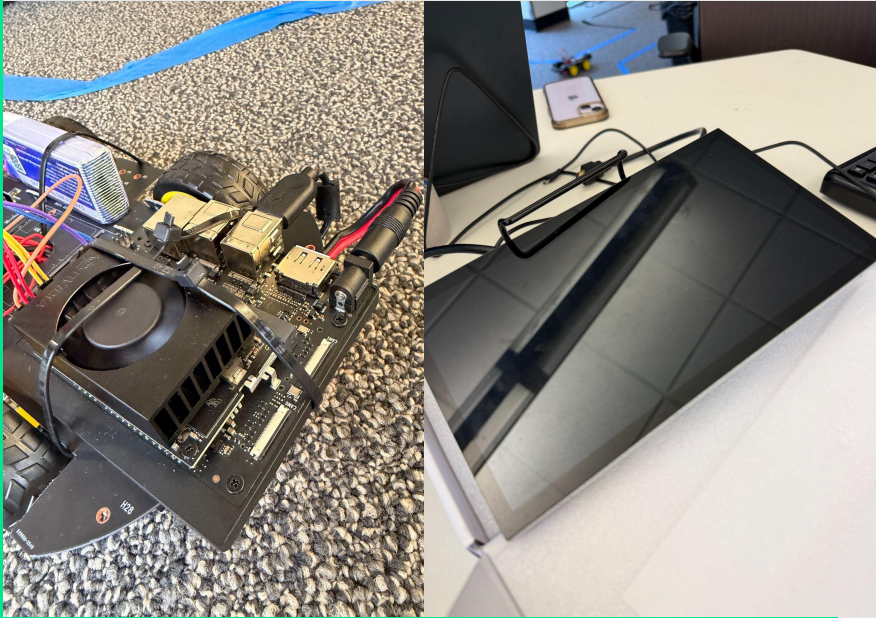
TensorRT - optimization

Flask, OpenCV, Pillow for utility/convenience

Training - Roboflow



Computer Vision/ Logic



- Jetson runs the computer vision algorithm and processes camera input while also controlling the motors of the RC car
- Computer vision algorithm will read camera input and differentiate between lane boundaries, and background
- Model trained beforehand with several hundred sets of obtained data
- Camera -> Segmentation -> Detection -> Steering

Demo

Self-Driving
Computer Vision
Project



- Detects lane boundaries
- Adjusts steering as it goes to stay inside lanes
- Minimal assistance
- Obstacle Avoidance

Results / Challenges / Future Improvements

1. CHALLENGES

- Construction / Wiring
- Training Consistency
- Debugging Software

2. RESULTS

- Successfully drove around the track without assistance
- Stayed within the lanes

3. IMPROVEMENTS

- More training data
- Improved hardware
- Increased consistency, especially in difficult environments